THE LAB — DAY 01

Welcome to the Lab

The Data Detective & Meet the Machines

Sunday · July 19, 2026 · Setup day · meet your (locked) AI Studio

1 Mission briefing

Welcome, recruit. Before we hand you a neural network, prove you can **read data** — every AI system is built on it, and a researcher who can't read their data ships broken models. This morning a crate of penguin measurements landed from an Antarctic field station: interrogate it. Then, as a reward, you'll play with four **finished** AI machines — the very ones you'll rebuild yourself over the next two weeks.

- [ ] Load a real dataset with **pandas** and size it up
- [ ] Hunt down **missing values** and clean the data
- [ ] **Filter, group, count** to answer real questions
- [ ] **Plot** the data and spot a learnable pattern
- [ ] See the **map of AI** (ML, deep learning, superpowers)
- [ ] Run four **pretrained** models — and try to fool them
- [ ] Meet the **AI Studio** you'll fill one tab per day

2 Field guide the concepts

AI, machine learning, deep learning aren't synonyms — they nest like Russian dolls. **Artificial intelligence** is any program we'd call smart; **machine learning** is the slice that isn't hand-coded but *learns patterns from data*; **deep learning** is machine learning with neural networks stacked many layers deep. There are three ways to learn: with labels (**supervised**), without (**unsupervised**), or by reward (**reinforcement**).

AI ⊃ machine learning ⊃ deep learning — and the three ways a machine can learn.

How ML data is shaped. Almost all of it is a table: one **row** per example (one penguin), one **column** per **feature** you measured (bill length, mass). Sometimes one column is the **label** — the answer you'll teach a machine to predict. Today's demos also use **pretrained** models: giant networks someone else trained, that you simply borrow.

> The Lab's motto: understand your data *before* you model it.

3 Lab missions check each off in the notebooks

NOTEBOOK 1A · THE DATA DETECTIVE

- [ ] **1 Count the gaps**
  Tally missing values per column with `penguins.isna().sum()`.
  Expect: 19 missing values in all — a few measurements, about a dozen unknown sexes.

- [ ] **2 The heavyweights**
  Filter for penguins heavier than 5000 g.
  Expect: about 60 birds — and nearly all the same species. Remember which.

- [ ] **3 The average penguin, by species**
  `groupby("species").mean(numeric_only=True)`.
  Expect: one row per species, Gentoo biggest on almost every measurement.

- [ ] **4 Who lives where?**
  Count penguins per island with `value_counts()`.
  Expect: Biscoe, Dream, Torgersen — Biscoe hosts the most.

- [ ] **5 The scatter plot that changes everything**
  Plot bill length vs bill depth, one color per species.
  Expect: three fairly distinct blobs — structure a machine could learn.

- [ ] **6 What moves together?**
  Correlation matrix of the four measurements, drawn as a heatmap.
  Expect: flipper length ↔ body mass is the brightest pair, about +0.87.

- [ ] **7 Close the case**
  Answer three questions with one line of pandas each.
  Expect: Gentoo (heaviest), Biscoe (most Gentoo), 195.8 mm (Chinstrap flipper).

NOTEBOOK 1B · MEET THE MACHINES

- [ ] **1 Fool the mood reader**
  Write a sarcastic line that *sounds* happy but means the opposite.
  Goal: get the sentiment model to answer POSITIVE for a clear complaint.

- [ ] **2 Be the author**
  Give GPT-2 your own opening line and generate.
  Expect: a different story each run (sampling is random).

- [ ] **3 Invent the categories**
  Hand the zero-shot sorter your own text and labels.
  Try: a sentence that could belong to two categories at once.

4 Cheat sheet the detective's toolkit

`pd.read_csv(path)`	Load a CSV file into a DataFrame
`df.head() · df.info() · df.describe()`	Peek at rows · column types & nulls · number stats
`df.isna().sum()`	Count missing values in each column
`df.dropna()`	Drop every row that has a missing value
`df[df["body_mass_g"] > 5000]`	Keep only the rows matching a condition
`df.groupby("species").mean(numeric_only=True)`	Average each numeric column per group
`df["island"].value_counts()`	Tally how often each value appears
`df[cols].corr()`	Correlation matrix (−1 … +1) of the columns
`plt.scatter(x, y, label=name)`	One dot per row; loop species to color them
`pipeline("text-classification", model=…)`	Load & run a pretrained model in one line

🔒 No module today — the padlock wall *is* the mission

Peek at the app you're about to build. From the repo root run:

```
$ python ai_studio/app.py
```

Every tab shows 🔒 — you haven't built anything yet. One notebook unlocks one tab: **Day 2** → 🐧 Prediction Machine, **Day 3** → 🎨 Pattern Finder, **Day 5** → ✍️ Digit Reader, and on to Demo Day, when every lock is gone.

6 Mini-glossary

dataset
A table you learn from — rows are examples, columns are what you measured.

feature
One input measurement (a column), e.g. bill length.

label
The answer you want to predict (e.g. species). Present in supervised data.

DataFrame
pandas' programmable spreadsheet — the `df` object.

pretrained model
A network already trained on huge data that you borrow as-is.

notebook kernel
The live Python process running your cells; restarting it wipes memory.

7 Tonight's show & tell

1. Share **one plot** you made and the single **insight** it revealed.
2. Which AI **superpower** impressed you most — and which one did you manage to **fool**?
3. What's the one **burning question** about AI you want answered by Demo Day?